Week 4 Lab 2 Report

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Task 1:

Define pins going into the board as the direction for down.

$$Voltage = \frac{Value}{1024} \times 3.3V$$

	Rest		Up		Down		Left		Right	
	Value	Volt(V)	Value	Volt(V)	Value	Volt(V)	Value	Volt	Value	Volt(V)
VRx	555	1.79	1024	3.30	9	0.03	555	1.79	555	1.79
VRy	542	1.75	542	1.75	542	1.75	9	0.03	1024	3.30

Task 2:

Bits	Val	Description
15	1	Start single conversion
14:12	100	$AIN_P = A0, AIN_N = GND$
11:9	000	$FSR = \pm 6.144\ V$
8	0	Continuous conversion mode
7:5	100	128 SPS (default)
4	0	Traditional comparator (default)
3	0	Active low (default)
2	0	Nonlatching. The ALERT/RDY
1:0	11	Disable comparator and set ALERT/RDY pin to high-impedance (default)

Bits 14:12 being 100 means we read from A0. To read the other analog-in pin, we simply change bits 14:12 to 101, so only bit 12 needs to be changed from 0 to 1.

Task 3:

Simply use thresholds to map the inputs to the appropriate state.

```
void direction(float v0, float v1) {
  byte isUp = v0 > HI_THRESH;
  byte isDown = v0 < LOW_THRESH;
  byte isLeft = v1 < LOW_THRESH;
  byte isRight = v1 > HI_THRESH;

  if (isUp) {
    if (isLeft)
      Serial.println("Up-Left");
    else if (isRight)
      Serial.println("Up-Right");
    else
      Serial.println("Up");
  } else if (isDown) {
```

```
    if (isLeft)
      Serial.println("Down-Left");
    else if (isRight)
      Serial.println("Down-Right");
    else
      Serial.println("Down");
  } else {
    if (isLeft)
      Serial.println("Left");
    else if (isRight)
      Serial.println("Right");
    else
      Serial.println("Rest");
  }
}

void click(float v2) {
  if (v2 < LOW_THRESH)
    Serial.println("Clicked!");
}
```

Task 4:

A new endpoint was added to change the state and write it to the LED pin.

```
void toggleLED() {
  LED_state = !LED_state;
  digitalWrite(LED, LED_state);
  if (LED_state)
    server.send(200, "text/plain", "LED off");
  else
    server.send(200, "text/plain", "LED on");
}
```

Task 5:

DHT Sensor from Lab 1 was used. A function was added to update the temperature and humidity readings, and the GET request is sent with the data in the URL, as follows:

```
url += "data?data1=" + String(temp) + "&data2=" + String(humid);
```